CS 5100 Final Project Proposal

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Kerbal Space Program (**KSP**) is a highly realistic physics-based space simulation game. Using the mod Kerbal Remote Procedure Call Server (**KRPC**), it is possible to control these vessels with Python, C++, or Lua scripts. The mod acts as an API and allows any 3rd-party library to be imported and used in scripts, for example, PyTorch or Pandas [1].

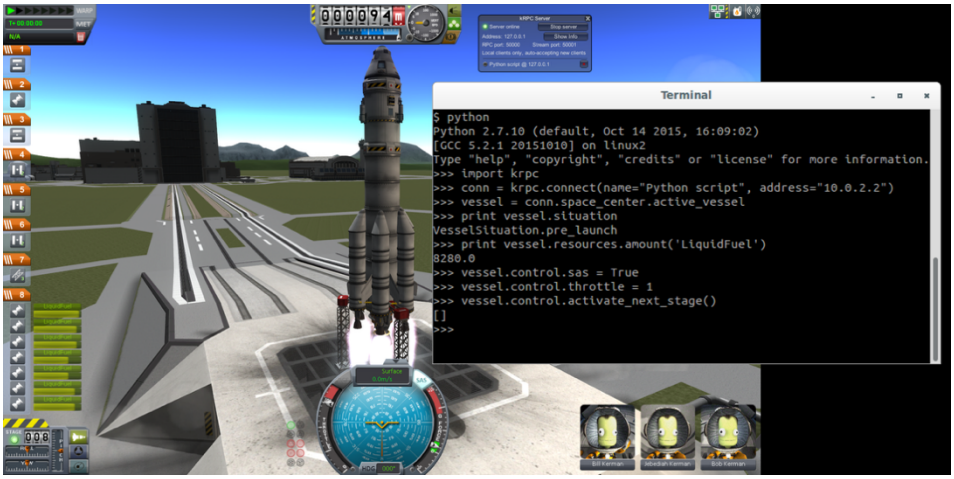

Figure 1: KSP space vessel being controlled by KRPC [2].

Once in-game, a server can be set up, and an agent program could then read from the vessel's sensors (altitude, longitude, latitude, speed, acceleration, fuel, pitch, yaw, temperature, stages). Then use all action groups in the game (lights, breaks, thrusters, deploy stages, rotors, levers…), and any custom action groups, such as deploying wings, landing legs, and more, as actuators.

There are many options here for the agent program; my interest mostly aligns with ML/RL and the latter topics of the course. Therefore, my project proposal is an ML model that enables an aircraft in the game to fly from point A to point B in the minimum time possible. Finishing with a plan to complete the route seen in Figure 2.

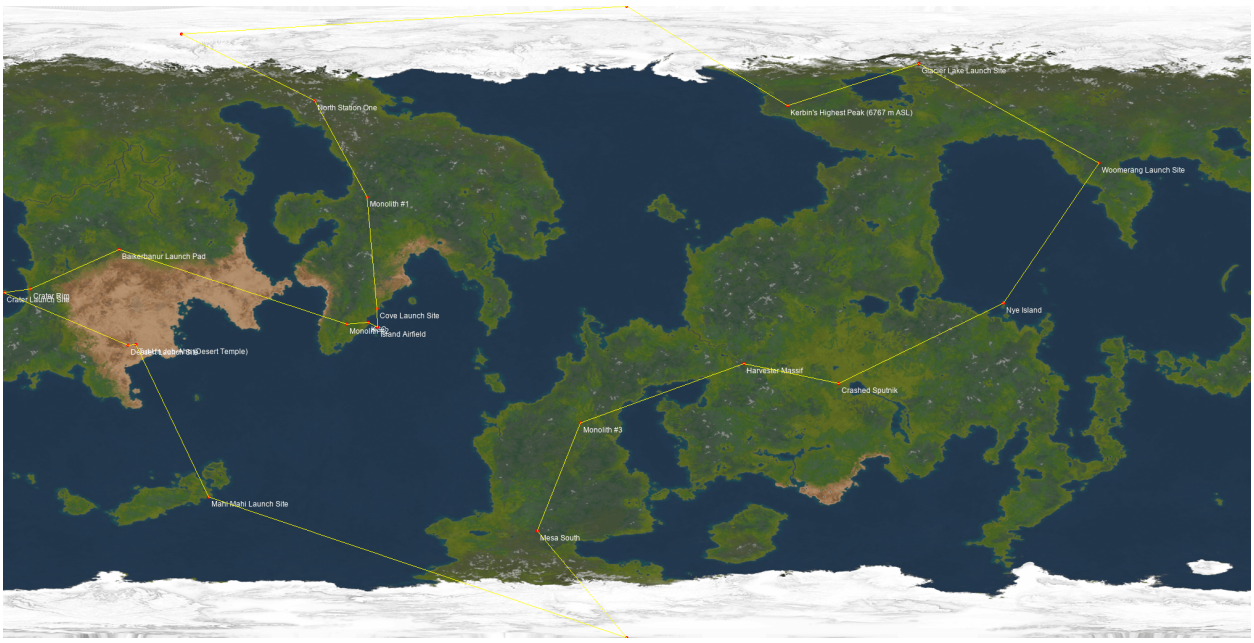

Figure 2: map of Kerbin (Earth analogue in KSP) with notable landmarks highlighted. [3]

The model would use reaching the target and time taken as goals, aiming to reach the target and minimize time taken. Note that, as time is minimized, more fuel is consumed, so the agent program must account for this in some way. Thankfully, aircraft can usually fly in a straight line, so the route will be easy to produce. Though landing and takeoff will be a significant challenge for the agent programs.

Bibliography

[1] "kRPC Documentation," kRPC Org, 2023. [Online]. Available: https://krpc.github.io/krpc/index.html. [Accessed 3 2 2026].

[2] KRPC, "KRPC Github," 10 7 2024. [Online]. Available: https://github.com/krpc/krpc. [Accessed 3 2 2026].

[3] NoUsernameSelected, "Spoiler warning] Here is the shortest path between every point of interest (launch site, comm

station, easter egg, etc.) on Kerbin:," 2026. [Online]. Available: https://www.google.com/url?sa=t&source=web&rct=j&url=https%3A%2F%2Fwww.reddit.com%2Fr%2FKerbalSpaceProgram%2Fcomments%2F1f8tszv%2Fspoiler_warning_here_is_the_shortest_path_between%2F&ved=0CBoQjhxqFwoTCKjVzb-bvpIDFQAAAAAdAAAAABAI&opi=89978449. [Accessed 3 2 2026].